

Seemab — Mathematics Exam Overview

SINGLE ATTEMPT

exam-13-ch1-2-4-5-7 (Version 1) | Units 1, 2, 4, 5, 7 | Federal Board Annual 2026
Date: 17 July 2026 | Total Marks: 65 | Time: 2 h 30 min | First attempt — fresh paper

58.5/65 TOTAL SCORE	90.0% PERCENTAGE	A+ GRADE	11/12 MCQ (91.7%)
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Section Breakdown

Section A (MCQ)

11/12	91.7%
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Section B (Short Q)

32/33	97.0%
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Section C (Long Q)

15.5/20	77.5%
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MCQ Results (11 of 12 Correct)

#	Topic	Key	Seemab	Result
1	Multiplicative inverse of 0 = undefined	D	D	✓
2	$5^{3/4}$ denominator = index of the radical	B	B	✓
3	Mantissa is always zero or positive	B	B	✓
4	$\log_2(-4)$ is undefined	C	C	✓
5	Reference position of 0.00532 = between 5 and 3	A	D	✗
6	HCF unchanged when scaled by same nonzero constant	A	A	✓
7	$x^2-11x+30 = (x-5)(x-6)$	A	A	✓
8	$\sqrt{49a^2-84ab+36b^2} = \pm(7a-6b)$	A	A	✓
9	$ x-4 =7 \rightarrow \{11,-3\}$	A	A	✓
10	$3(-2) > 5(-2)$ when multiplied by negative c	B	B	✓
11	(3,-5) lies in the Fourth quadrant	D	D	✓
12	P midpoint of RS, PS=6 $\Rightarrow$ RS=12	C	C	✓

One miss (Q5): "reference position" terminology — correct answer sits right after the first significant digit, not at the decimal point. Math MCQ all-time now 106/108 (98.1%; 2 errors: exam-07, exam-13).

Per-Question Summary

Question	Topic	Marks	Earned	Notes
Q2(i)	Scientific notation (distance)	3	3	4.2×10^1 km correct
Q2(ii)	Temperature scale conversion	3	3	-128.56°F and 183.8 K both correct
Q2(iii)	Equality/inequality properties	3	3	Transitive + multiplicative, both correct
Q2(iv)	Antilogarithm	3	3	270.7 and 38.82 — more precise than the key's own 270.8
Q2(v)	Domain of a logarithm	3	3	All 4 sub-cases correct
Q2(vii)	Sum/difference of cubes & squares	3	2.5	Correct working but final line dropped the x^2 (−0.5)
Q2(viii)	HCF by factorization	3	3	HCF=(3x+1) correct
Q2(ix)	Square root by factorization	3	2.5	Correct factoring but final line kept a stray exponent, $\pm(5a^2-3a)$ never written (−0.5)
Q2(xi)	Checking inequality solutions	3	3	Both sub-parts verified correctly
Q2(xii)	Absolute value equation	3	3	$\{5/2, -2/3\}$ correct
Q2(xiii)	Midpoint formula (decimals)	3	3	(2,1) correct
Q3(i)	Trichotomy + ratio word problem	5	3.5	Trichotomy correct (2/2); ratio method correct but total penalty misread $10\times$ too large (1.5/3)
Q3(ii)	Log laws + digit count	5	3.5	$\log(28/45)$ correct (3/3); $40 \times \log 7$ multiplication slip gave 35 digits not 34 (0.5/2)
Q3(iv)	Compound inequality + abs. value	5	4.5	Number line correct; case-2 direction flipped without justification (−0.5)
Q3(v)	Midpoint quadrilateral (Varignon)	5	4	All midpoints & sides correct; quadrilateral classification never written (−1)

Headline: a solid paper at 58.5/65 (90.0%, A+) with a strong (though not perfect) Section B at 32/33 — 9 of the 11 attempted short questions scored full marks. The 6.5 lost marks are spread across all three sections: one reference-position MCQ term, two final-line transcription slips in Section B (a dropped x^2 , a stray exponent), and in Section C a $10\times$ misread of a given Rupee figure, an arithmetic slip in a $\log\times 40$ multiplication, an inequality case that flipped direction without justification, and a quadrilateral-type

conclusion that was never written despite all the supporting numeric work being correct.

Performance Strengths

- Ratio method (1:2:3 split) set up and applied correctly despite the misread penalty figure
- Logarithm work strong: $\log(28/45)$ via log laws correct; $\text{antilog}(2.4324)$ beats the answer key's own stated value
- Varignon midpoint quadrilateral: all four midpoints AND all four side lengths ($\sqrt{37}/2$, 4, $\sqrt{37}/2$, 4) correct
- Number-line plotting done properly — solution circles at -2 and 7 confirmed on deep-check
- MCQ consistency at 11/12, in line with Seemab's established baseline

Areas to Improve

- **Final-line transcription care:** three losses this exam are "working correct, final line wrong or missing" — the dropped x^2 in Q2(vii)(a), the stray exponent in Q2(ix)'s root, and the unwritten quadrilateral classification in Q3(v)(c)
- **Inequality-direction consistency:** Q3(iv)(a)'s case 2 flipped sign mid-working with no justification shown
- **Digit-level arithmetic care:** 40×0.8451 computed as 34.04 instead of 33.804, flipping the digit-count answer
- **Read given numbers carefully:** Rs. 600,000 misread as Rs. 6,000,000 in the ratio word problem, despite a fully correct method
- **Recurring pattern (3rd occurrence):** the missing final naming/classifying/concluding sentence has now shown up on exam-10, exam-12, and exam-13

Math Subject Performance History

#	Date	Exam	Score	%	MCQ	Sec B	Sec C
1	18 Apr 2026	exam-01-ch1-2	65/65	100%	12/12	33/33	20/20
2	22 Apr 2026	exam-03-ch4-5-7	60.5/65	93.1%	12/12	29.5/33	19/20
3	25 May 2026	exam-04-ch4	57/65	87.7%	12/12	26/33	19/20
4	5 Jun 2026	exam-07-ch8	64/65	98.5%	11/12	33/33	20/20
5	26 Jun 2026	exam-10-ch1-2-4-5-7-10	61/65	93.8%	12/12	32/33	17/20
6	3 Jul 2026	exam-11-ch1-2-4-5-7 (1st attempt)	58.5/65	90.0%	12/12	31.5/33	15/20
7	5 Jul 2026	exam-11-ch1-2-4-5-7 (2nd attempt)	63.5/65	97.7%	12/12	32/33	19.5/20
8	9 Jul 2026	exam-12-ch1-2-4-5-7	62/65	95.4%	12/12	32/33	18.5/20
9	17 Jul 2026	exam-13-ch1-2-4-5-7 NEW	58.5/65	90.0%	11/12	32/33	15.5/20

Subject average after this attempt: $550.0/585 = \mathbf{94.0\%}$ (down from 94.5%). MCQ all-time: 106/108 (98.1%; 2 errors: exam-07, exam-13). Math now sits in a dead heat with Chemistry at 94.0%, both trailing Physics (95.0%).

Note for Parent

This was a solid result: 58.5/65, A+, on a paper Seemab had already sat a very similar version of before (exam-12, 62/65). Section B was strong but not perfect at 32/33 — 9 of the 11 short questions she chose scored full marks, including harder items like the antilog and HCF-by-factorization questions, while two (the cubes factorization and a square root) lost half a mark each for dropping or garbling the very last line of otherwise-correct working. The remaining marks lost were in Section C and MCQ, and again came mostly from small slips rather than a lack of understanding: she read "Rs. 600,000" as "Rs. 6,000,000" in one word problem (the method was completely correct, just applied to the wrong number), made a multiplication slip in a logarithm digit-count question, flipped an inequality's direction mid-working without showing why, and left out the final "so EFGH is a parallelogram" sentence on a coordinate-geometry question even though she had already worked out all the numbers needed to say it. None of these are conceptual gaps — they are all careful-reading and always-finish-the-sentence habits that should be quick to fix with a bit of practice.